



Figure 1: High level workflow

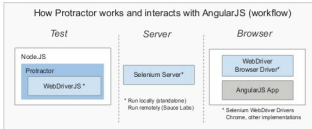


Figure 2: Interaction of Protractor with AngularJS

Prerequisites

Protractor has no prerequisites except for the knowledge of JavaScript, which plays a very significant role in Web development because it's the only language browsers understand. If you already know Angular, that is an added advantage.

Installation

The setup is very easy and straightforward. Here is a list of what you need to install.

1. Node.js
2. Protractor

Step 1: Installing Node.js

- Go to <https://www.npmjs.com/> and download *npm* which comes with Node.js.
- Install *npm* in your system.
- After completing the installation, use the command prompt to enter the command:

```
npm -v.
```

If *npm* is installed successfully, this command will give you the version of *npm*.

- To check the version of Node, enter the command:

```
node -version or node -v.
```

Step 2: Installing Protractor

- Open the command prompt and enter the command:

```
npm install -g protractor.
```

- This will install Protractor with Jasmine and the *webdriver-manager*. By default, Protractor uses the

```

Administrator: C:\Windows\System32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

D:\>npm -v
5.2.0
D:\>node -v
v8.2.1
D:\>node --version
v8.2.1
D:\>
  
```

Figure 3: Installation

Jasmine test framework for its testing interface. The *webdriver-manager* is a helper tool to easily get an instance of a running Selenium server

- After Protractor is installed, update the *webdriver-manager* with the command:

```
webdriver-manager update
```

...and then start it by using the command:

```
webdriver-manager start.
```

Using Protractor for AngularJS

Let's start by creating a new folder with any name or location of your choice. In that folder, create the following two files:

```
conf.js
spec.js
```

spec.js and *conf.js* will store the specifications and configurations respectively, for execution.

conf.js: This is the file in which we define/configure the details of the framework, Selenium, specs, script timeouts, *Onprepare()* functions, capabilities, and reports. As soon as *Run* starts, Protractor looks for this file first and tries to execute the code written in that file. It specifies that we should use Jasmine for the test framework. It will use the defaults for all other configurations. Chrome is the default browser.

Your *conf.js* file will contain something like what follows:

```

exports.config = {
  framework: 'jasmine',
  seleniumAddress: 'http://localhost:4444/wd/hub',
  specs: ['spec.js']
}
  
```

spec.js: *spec.js* is a file in which you define how Jasmine, a behaviour-driven development framework for testing JavaScript code, works. *spec.js* mainly contains two functions: *describe()* and *it()*. *describe()* is used to describe an action. *It()* simply contain actual tests with logic to run.